

Where the work runs

Host memory

the consensus state

one input per agent



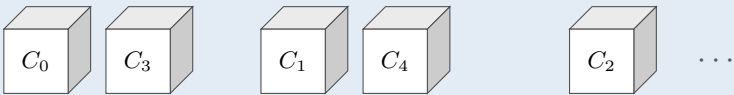
Each finished task's output region is folded into the state under a lock, so no device holds the 4D arrays.

Key

-  data fit, fixed per device
-  denoising, shared queue
-  the consensus state on the host

Data-fit tasks

one per frame of the 4D image



Frame t runs on device $t \bmod n$, with n the size of the pool, and the choice is fixed for the run. The frame's slab of the sinogram y_t is placed on that device once.

fold the output region



One worker thread per device runs both kinds of task.

`set_device_pool` takes `None` (every GPU, or the CPU when there is none), `'cpu'` or `'gpu'`, a count, or a list of device indices.

`MBIRTORCH_NUM_DEVICES` pins the count for a whole process.

Denoise tasks



Each orientation is cut into slabs of hyperplane volumes sized by `denoise_slab_gb` (default 2.0 GB). The slabs go into one shared queue, and any device takes the next slab when it is free.